Test Plan: VWO Login Dashboard

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1. Objective

The goal of this test plan is to ensure the quality, functionality, security, accessibility, and reliability of the VWO Login Dashboard hosted at app.vwo.com. The login dashboard is the critical entry point for users accessing VWO's experimentation, personalization, and analytics tools, and it serves digital marketers, product managers, UX designers, developers, enterprise teams, conversion rate optimization specialists, and data analysts.

This test plan verifies that the login experience:

- Provides secure access to VWO's experimentation platform with enterprise-grade security standards.
- Minimizes login friction to improve user adoption and retention.
- Supports enterprise security requirements and compliance standards (GDPR, CCPA, OWASP authentication guidelines, WCAG 2.1 AA).
- Facilitates seamless onboarding for new users discovering VWO's capabilities.
- Meets the stated success metrics: 95%+ login success rate, sub-2-second page load, 90%+ user satisfaction, zero successful brute force attacks, and a 20% reduction in login-related support tickets.

2. Scope

Scope of the Test Plan for the VWO Login Dashboard:

1. **Functional Testing:**
    - Verify email and password based login with secure validation.
    - Test the Remember Me option for persistent login sessions.
    - Validate the account registration link and free trial signup path for new users.
    - Validate the transition to the main VWO dashboard after successful authentication.
2. **User Input Validation Testing:**
    - Verify real-time field validation on blur with immediate feedback.
    - Validate email format verification and specialized mobile keyboards.
    - Validate password strength indicators and password requirement feedback.
    - Test boundary values for input fields to check for unexpected behavior.
3. **Error Handling Testing:**
    - Verify clear, actionable error messages for failed authentication attempts.
    - Check error responses for sensitive information disclosure.
    - Validate error identification, recovery options, and success confirmation in the error recovery flow.
4. **Password Management Testing:**
    - Verify the forgot password flow with secure token generation.
    - Test email-based password recovery and multiple recovery options.
    - Validate enforced password complexity requirements.
5. **Session Management Testing:**
    - Verify secure session handling with configurable timeout periods.
    - Validate secure session token generation and management.
    - Test session behavior with and without Remember Me enabled.

6. **Multi-Factor and Enterprise Authentication Testing:**
    - Verify optional 2FA support for enhanced security.
    - Validate enterprise SSO integration for organizational accounts.
    - Test SAML, OAuth, and other enterprise authentication protocols.
    - Test optional social login with Google, Microsoft, and other identity providers.
7. **Security Testing:**
    - Validate end-to-end encryption for all authentication data transmission.
    - Verify encrypted password storage using industry-standard hashing algorithms.
    - Verify HTTPS enforcement and SSL/TLS encryption for all login communications.
    - Validate rate limiting and request throttling protection against brute force attacks.
    - Conduct security assessments for vulnerabilities such as SQL injection and XSS.
8. **Performance Testing:**
    - Verify login page loading within 2 seconds on standard connections.
    - Validate compressed images and minified CSS/JavaScript assets.
    - Verify CDN utilization for global performance.
    - Assess response time under normal and peak loads to identify bottlenecks.
9. **Scalability and Availability Testing:**
    - Validate 99.9% uptime expectations for the global user base.
    - Test support for thousands of simultaneous login attempts.
    - Validate multi-region deployment for optimal global performance.
10. **Usability and Interface Testing:**
    - Verify responsive design and mobile-optimized touch-friendly controls.
    - Verify auto-focus on the first input field and clickable form labels.
    - Validate loading states and clear feedback during authentication processing.
    - Validate Light and Dark Mode theme support.
11. **Accessibility Testing:**
    - Verify screen reader support with ARIA labels.
    - Verify full keyboard navigation for all interactive elements.
    - Validate high contrast mode for visually impaired users.
    - Validate WCAG 2.1 AA compliance and inclusive design principles.
12. **Branding and Visual Design Testing:**
    - Verify brand consistency with VWO's design system and color scheme.
    - Validate professional, trustworthy visual appearance.
    - Verify the product announcement banner for the new UI launch.
13. **Integration Testing:**
    - Verify seamless transition to the VWO core platform after authentication.
    - Validate login success and failure tracking for analytics integration.
    - Validate integration with customer support systems for login assistance.
    - Validate marketing tool integration with onboarding and analytics platforms.
14. **Compliance Testing:**

- Validate GDPR and CCPA compliance for user data handling.
- Validate support for enterprise security policies and audit trails.
- Verify adherence to OWASP authentication guidelines.

15. **Compatibility Testing:**
    - Test the login dashboard across supported browsers, operating systems, and devices.
16. **Regression Testing:**
    - Conduct regression testing after bug fixes or updates to ensure existing functionality remains intact.
17. **Edge Case and Ad Hoc Testing:**
    - Test extreme and boundary scenarios to identify potential issues.
    - Perform exploratory testing to identify hidden defects or usability issues.
18. **Concurrency Testing:**
    - Assess behavior when multiple users attempt to log in simultaneously.
19. **Continuous Integration and Deployment (CI/CD) Testing:**
    - Validate login dashboard behavior within the CI/CD pipeline to ensure smooth deployments.
20. **Monitoring and Analytics Testing:**
    - Validate real-time performance and security monitoring and alerting.
    - Validate login pattern analytics and personalization inputs.

It is important to note that the scope of this test plan may evolve during the testing process based on feedback, changing requirements, or discoveries during testing. The scope should be reviewed and adjusted accordingly throughout the testing phase to ensure comprehensive coverage of the VWO Login Dashboard.

3. Inclusions

Login Operations

- Test successful login with valid registered email and password.
- Verify appropriate error responses for invalid credentials, unregistered email, and empty mandatory fields.
- Validate that a successful login establishes a secure session and redirects to the personalized VWO dashboard.
- Verify the Remember Me checkbox creates a persistent login session and that unchecking it does not.

Registration and New User Path

- Test the account registration link and free trial signup call-to-action with minimal friction.
- Verify the guided onboarding introduction to VWO's capabilities post-registration.
- Validate discovery entry points from VWO marketing materials and referrals.

Password Management

- Test the forgot password flow end to end, including secure token generation.
- Verify email-based password reset and alternate recovery options.
- Validate password complexity enforcement and password strength indicators.
- Verify reset tokens expire and cannot be reused.

Input Validation

- Test validation on blur for email and password fields.
- Verify email format validation and mobile keyboard type for the email field.
- Test boundary values for minimum and maximum allowed input lengths.
- Verify handling of invalid characters, invalid data types, and mandatory field omissions.

Session and Authentication Controls

- Test session timeout behavior against the configured timeout period.
- Verify secure session token generation and management.
- Test optional 2FA enablement, verification, and recovery.
- Test enterprise SSO login for organizational accounts, including SAML and OAuth.
- Test optional social login with Google, Microsoft, and other identity providers.

Security Validation

- Verify HTTPS enforcement and SSL/TLS encryption on all login communications.
- Verify authentication data is encrypted end to end in transmission.
- Verify passwords are stored using industry-standard hashing algorithms.
- Test rate limiting and throttling against repeated failed login attempts.
- Test for SQL injection, XSS, and sensitive information disclosure in error responses.
- Verify no unauthorized session hijacking is possible.

User Experience and Accessibility

- Verify auto-focus on the first input field and clickable labels.
- Verify loading states during authentication processing.
- Verify responsive layout and touch-friendly controls on mobile devices.
- Verify screen reader support, ARIA labels, and full keyboard navigation.
- Verify high contrast mode and Light and Dark Mode theme support.
- Verify brand consistency and the new UI announcement banner.

Performance and Scalability

- Measure login page load time against the sub-2-second target on standard connections.
- Verify asset optimization through compressed images and minified CSS/JavaScript.
- Verify CDN delivery and multi-region performance.
- Evaluate behavior under high concurrent login loads to confirm stability.

Integration and Analytics

- Verify seamless transition to the VWO core platform after successful authentication.
- Verify login success and failure events are tracked for platform optimization.
- Verify integration with customer support systems for login assistance.
- Verify context preservation and recent activity from previous sessions.

Regression and Exploratory

- Perform regression tests after bug fixes or updates to ensure existing login functionality remains intact.
- Perform exploratory testing to surface hidden defects and usability issues.

4. Test Environments

The following operating systems, browsers, devices, and network conditions will be used for testing.

Environment URLs

Name	Environment URL	Purpose
QA	https://app.vwo.com/login	Functional, UI, accessibility, and regression testing
Pre Prod	https://app.vwo.com/login	UAT, performance, security, and release validation

Operating Systems and Browsers

- Windows 10: Chrome, Firefox, and Edge
- macOS: Safari Browser
- Android Mobile OS: Chrome
- iPhone Mobile OS: Safari

Devices and Screen Sizes

- Desktop computers and laptops
- Tablets
- Smartphones with touch-friendly control validation

Network Connectivity

- Wi-Fi, cellular, and wired connections
- Standard connection profile for the sub-2-second page load target
- Throttled and high-latency profiles for global and CDN performance validation

Hardware and Software Requirements

- Sufficient processor, memory, and storage capacity to run the browser matrix and load generation tooling concurrently
- Screen reader software and keyboard-only navigation setup for accessibility testing
- Load testing tooling capable of simulating thousands of simultaneous login attempts

Security Protocols and Authentication

- HTTPS with SSL/TLS for all test environment access
- Test accounts for standard, 2FA-enabled, SSO, and social login scenarios
- Tokens or certificates as required for enterprise SSO test tenants

Access Permissions and Roles

- Testers: execute test cases, raise defects, and update status
- Developers: triage and fix defects, provide builds
- Stakeholders: review reports and provide approvals

5. Defect Reporting Procedure

- **Defect identification criteria:** deviation from the requirements in the PRD, user experience issues, accessibility non-compliance, security weaknesses, performance regressions, or technical errors.
- **Reporting steps:** use the designated defect template, provide detailed reproduction steps, attach screenshots or logs, and record the environment, browser, and device used.
- **Triage and prioritization:** assign severity and priority levels, then assign the defect to the appropriate team member for investigation and resolution.
- **Tracking tools:** defects are tracked and managed in JIRA.
- **Roles and responsibilities:** testers raise and verify defects, developers fix them, and the test lead owns triage, prioritization, and reporting.
- **Communication:** a daily end-of-day status email is sent to development and management listing the defects found that day.
- **Metrics:** number of defects found, time taken to resolve, percentage of defects successfully fixed, defect severity distribution, and defect leakage into later phases.

Defect Process Points of Contact

Area	Point of Contact
New Frontend	Devesh
Backend	Sonal
Dev Ops	Prajeeth

6. Test Strategy

Step 1: Test Design

The first step is to create test scenarios and test cases for the various features in scope.

While developing test cases, we will use a number of test design techniques:

- Equivalence Class Partition
- Boundary Value Analysis
- Decision Table Testing
- State Transition Testing
- Use Case Testing

We also use our expertise in creating test cases by applying the below:

- Error Guessing
- Exploratory Testing

We prioritize the test cases.

Step 2: Testing Procedure When We Receive a Request for Testing

- First, we conduct smoke testing to see if the various and important functionalities of the application are working.
- We reject the build if the smoke testing fails and will wait for the stable build before performing in-depth testing of the application functionalities.
- Once we receive a stable build which passes smoke testing, we perform in-depth testing using the test cases created.
- Multiple test resources will be testing the same application on multiple supported environments simultaneously.
- We then report the bugs in the bug tracking tool and send development and management the defects found on that day in a status end-of-day email.

As part of the testing, we will perform the below types of testing:

- Smoke Testing and Sanity Testing
- Regression Testing and Retesting
- Usability Testing, Functionality and UI Testing
- Security Testing, Accessibility Testing, and Performance Testing

We repeat test cycles until we get the quality product.

Step 3: Best Practices

- **Context Driven Testing:** we will be performing testing as per the context of the given application.
- **Shift Left Testing:** we will start testing from the beginning stages of the development itself, instead of waiting for the stable build.
- **Exploratory Testing:** using our expertise we will perform exploratory testing, apart from the normal execution of the test cases.
- **End to End Flow Testing:** we will test the end-to-end scenario which involves multiple functionalities to simulate the end user flows, including new user registration, returning user login, and error recovery.

7. Test Schedule

Following is the test schedule planned for the project. Two sprints are planned to test the application.

Task	Dates	Duration
Creating Test Plan	Sprint 1, Week 1	2 days
Test Case Creation	Sprint 1, Week 1 to Week 2	5 days
Test Case Execution	Sprint 1, Week 2 to Sprint 2, Week 2	2 sprints
Summary Reports Submission Date	Sprint 2, Week 2	2 days

> Phase alignment: Phase 1 covers core authentication (secure login form, basic validation and error handling, password reset). Phase 2 covers enhanced UX (mobile optimization, responsive design, accessibility, advanced validation and feedback). Phase 3 covers enterprise features (SSO integration, advanced security, analytics and monitoring).

8. Test Deliverables

Deliverable	Description	Owner
Test Plan	This document, covering objective, scope, strategy, schedule, and risks	QA Team
Test Scenarios	High-level scenarios derived from the PRD requirements	QA Team
Test Cases	Detailed test cases with steps, test data, and expected results	QA Team
Requirement Traceability Matrix	Mapping of PRD requirements to test scenarios and test cases	QA Team
Test Execution Report	Pass, fail, and blocked status per test cycle	QA Team
Defect Report	Defects logged in JIRA with severity, priority, and status	QA Team
Test Summary Report	Consolidated results, metrics, and release recommendation	Test Lead
Accessibility and Performance Reports	WCAG 2.1 AA findings and load test results	QA Team

9. Entry and Exit Criteria

The below are the entry and exit criteria for every phase of the Software Testing Life Cycle.

Requirement Analysis

Entry Criteria:

- Once the testing team receives the requirements documents or details about the project.

Exit Criteria:

- List of requirements are explored and understood by the testing team.
- Doubts are cleared.

10. Test Execution

Entry Criteria:

- Test scenarios and test cases documents are signed off by the client.
- Application is ready for testing.

Exit Criteria:

- Test case reports and defect reports are ready.

11. Test Closure

Entry Criteria:

- Test case reports and defect reports are ready.

Exit Criteria:

- Test summary reports are published and approved.

12. Tools

The following are the list of tools we will be using in this project:

- JIRA Bug Tracking Tool
- Mind map Tool
- Snipping Screenshot Tool
- Word and Excel documents
- Browser developer tools for network, console, and responsive checks
- Screen reader and accessibility audit tooling for WCAG 2.1 AA validation
- Load and performance testing tooling for concurrent login simulation

13. Risks and Mitigations

The following are the list of risks possible and the ways to mitigate them:

Risk	Mitigation
Non-availability of a resource	Backup resource planning
Build URL is not working	Resources will work on other tasks
Less time for testing	Ramp up the resources based on the client needs dynamically
Security vulnerabilities such as brute force, SQL injection, or XSS	Regular security audits and penetration testing, real-time security monitoring and alert systems, and regular security patch deployment and vulnerability assessments
Performance degradation under peak login load	Comprehensive load testing under various load conditions, real-time performance monitoring and alerting, and auto-scaling infrastructure to handle traffic spikes
Accessibility non-compliance against WCAG 2.1 AA	Regular accessibility testing and incorporation of user feedback
Compliance gaps for GDPR, CCPA, or enterprise audit requirements	Compliance review of user data handling and audit trail validation in every test cycle

14. Approvals

Team will send different types of documents for client approval like below:

- Test Plan
- Test Scenarios
- Test Cases
- Reports

Testing will only continue to the next steps once these approvals are done.

Role	Name	Signature	Date
Test Lead			
QA Manager			
Development Lead			
Product Owner			
